

EC1208

Principles of the Macro Economy

Summer 2022 Examination

Paper 1 · 1.5 hours · Section A: answer 2 of 3 (40%) · Section B: answer 1 of 2 (60%)

Module	EC1208 — Principles of the Macro Economy
Session	Summer 2022
Structure	Section A: three short questions (equal weight) — answer any two, together worth 40% of the total. Section B: two long questions with equally-weighted parts — answer any one, worth 60% of the total.
This scheme covers	All three Section A questions AND both Section B questions in full, so you can study every possible combination rather than just the ones you'd pick on the day.

A note on reused content. Question B1 on this paper is identical to Question B1 on the EC1208 Summer 2023 paper (productivity, the Solow model, Malthus, and institutions) — the model answer here is the same verified content, since the questions are worded identically.

Section A — Short Questions

Answer 2 of 3 · 40% of total marks · each question worth 100% of its own mark allocation

Question A1

100%

Using the 19th century neoclassical theory of distribution, based on the marginal productivity of each factor of production, determine what happened to both (i) and (ii) below when the Black Death hit Europe in 1348.

(i) The real wage.

(ii) The real rental price of capital.

FULL MODEL ANSWER

Under the 19th century neoclassical theory of distribution, each factor of production is paid its marginal product — the extra output generated by one additional unit of that factor, holding all other factors fixed. The Black Death of 1348 was, in economic terms, an enormous negative labor supply shock: it is estimated to have killed somewhere between a quarter and a half of Europe's population within a few years, while the stock of land and capital survived largely intact.

(i) The Real Wage: Rose

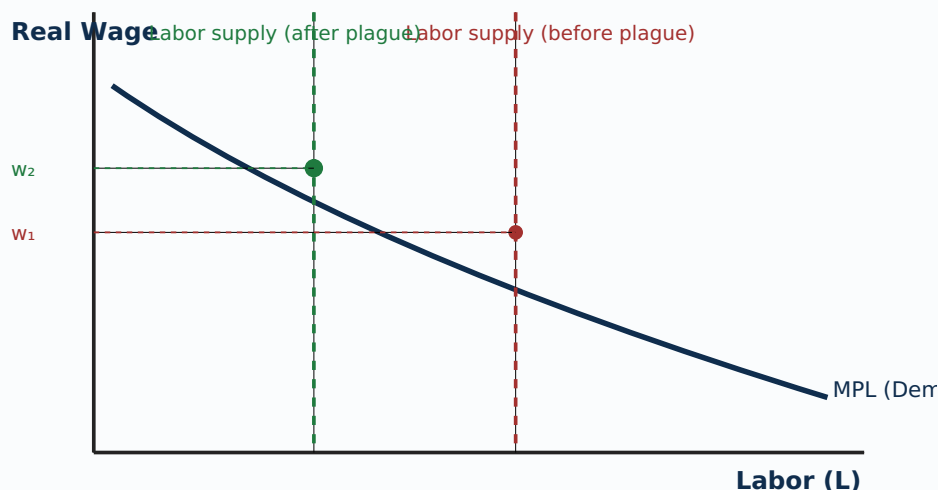


Figure 1: The labor market. A sharp fall in the labor supply (population collapse) moves the market up the downward-sloping marginal product of labor (MPL) curve, raising the equilibrium real wage from w_1 to w_2 .

The mechanism: a smaller workforce combined with an unchanged capital/land stock raises the marginal product of labor. With substantially fewer workers available but essentially the same amount of land and capital to work with, each remaining worker now had more land and capital to combine their labor with than before — and under diminishing marginal product, less labor relative to a fixed factor means each unit of labor becomes more productive at the margin. Since neoclassical distribution theory pays labor its marginal product, the real wage rose as the labor supply collapsed, moving the labor market up along a downward-sloping marginal product of labor curve (Figure 1).

This is well documented historically, once the initial disruption settled.

Land rents fell and real wages rose substantially across most of Europe in the century or so following the plague — in England, real wages roughly doubled over the following century, and the period is sometimes described as a "Golden Age" for labourers, precisely consistent with the theoretical prediction. There is a genuine, well-documented nuance here: in the immediate one to two years after 1348, severe supply disruption and a burst of price inflation temporarily depressed real wages in some regions even as nominal wages rose — but the underlying marginal-productivity mechanism reasserted itself clearly within a few decades, and the sustained medium- to long-run rise in real wages is the standard, well-supported result.

(ii) The Real Rental Price of Capital (Land): Fell

The mechanism: the same logic applies in reverse to capital/land. Land and capital were now relatively abundant compared to the sharply reduced workforce available to work them — each unit of land or capital now had fewer workers combining with it than before. Under diminishing marginal product, more of a factor relative to a fixed complementary factor (here, labor) means each additional unit of that factor is less productive at the margin. The marginal product of land and capital therefore fell, and since neoclassical theory pays capital its marginal product, the real rental price of capital/land fell as well.

This is also well documented historically. Land rents declined substantially across Western Europe in the century following the Black Death, and landlords were forced to offer increasingly favourable tenancy terms to attract scarce tenant labour — the mirror image of the rising real wage, and exactly what the marginal productivity theory predicts when labor becomes scarce relative to land and capital.

A complete answer explicitly connects both outcomes to the same underlying mechanism: labor became scarce relative to capital/land, raising labor's marginal product (and wage) while lowering capital/land's marginal product (and rental price) — the two results are two sides of the exact same relative-scarcity shift, not two separate phenomena to explain independently.

How the 100% Is Likely Allocated

Tier	Marks	What separates this tier
80-100%	80-100%	Correctly applies marginal productivity theory to both factors, explains the mechanism (relative scarcity of labor vs. capital/land shifting each factor's marginal product in opposite directions) for both, and connects the theoretical prediction to the historical record (rising wages, falling rents).
50-79%	50-79%	Correctly identifies both directional outcomes (wages up, rents down) but with less complete explanation of the underlying marginal product mechanism.
20-49%	20-49%	Correctly identifies one outcome but not the other, or identifies both directions without applying the marginal productivity framework explicitly.
0-19%	0-19%	No meaningful attempt, or incorrect direction for both.

Question A2**100%**

Explain, with the aid of diagram(s), the impact of a fiscal contraction by a large nation such as the United States on a "small open economy", for the following:

- (i) The trade balance.
- (ii) The real exchange rate.

FULL MODEL ANSWER

A small open economy is a price-taker in world capital markets: it takes the prevailing world real interest rate, r^* , as given, since its own saving and investment decisions are too small to meaningfully move the global rate. A fiscal contraction by a large economy like the United States — a smaller budget deficit or a surplus — increases that large economy's own national saving. Because the U.S. is large enough to influence world capital markets, this raises the global supply of loanable funds and pushes the **world real interest rate down**. That single change then works through the small open economy in two connected steps.

Step 1: The Fall in the World Interest Rate Reduces Net Capital Outflow

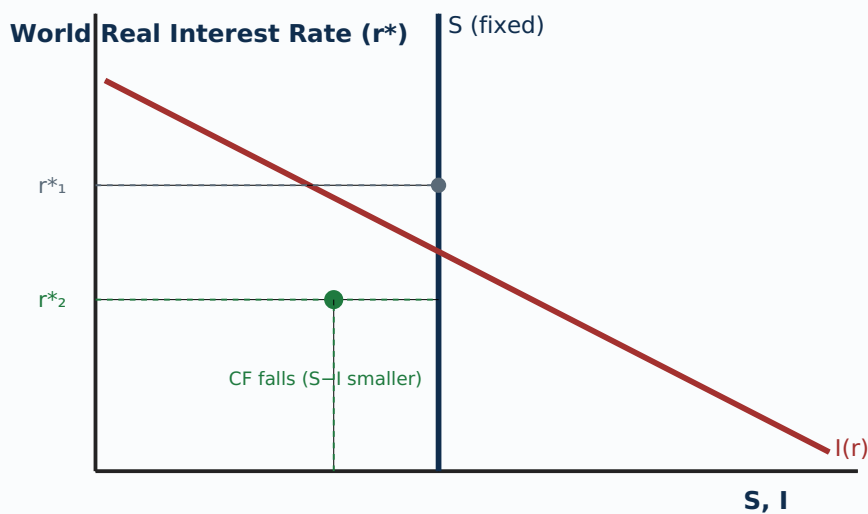


Figure 1: The small open economy's loanable funds market. National saving (S) is fixed (vertical); investment $I(r)$ is downward-sloping. A fall in the world interest rate from r^*_1 to r^*_2 (driven by the large nation's fiscal contraction) raises investment, so net capital outflow ($CF = S - I$, the gap between S and I) shrinks.

The mechanism. In the small open economy's own loanable funds market, national saving (S) is essentially fixed (unaffected by the world interest rate in this simple framework), while domestic investment, $I(r)$, is downward-sloping — lower interest rates make more investment projects profitable. As the world interest rate falls from r^*_1 to r^*_2 , domestic investment rises along the unchanged $I(r)$ curve. Since net capital outflow is defined as $CF = S - I$, and S is unchanged while I rises, CF falls.

(i) Effect on the Trade Balance: Worsens (Falls)

The national accounting identity linking CF and NX. In the balance of payments, net capital outflow must equal net exports: $CF = NX$. This is an accounting identity, not a behavioural relationship — every euro (or dollar) of goods a country exports without a matching import must be financed by an equivalent net outflow of capital abroad, and vice versa. Since CF falls (Step 1), NX must fall by exactly the same amount — the small open economy's **trade balance worsens** (moves toward deficit, or a smaller surplus) as a direct consequence of the large nation's fiscal contraction.

(ii) Effect on the Real Exchange Rate: Appreciates

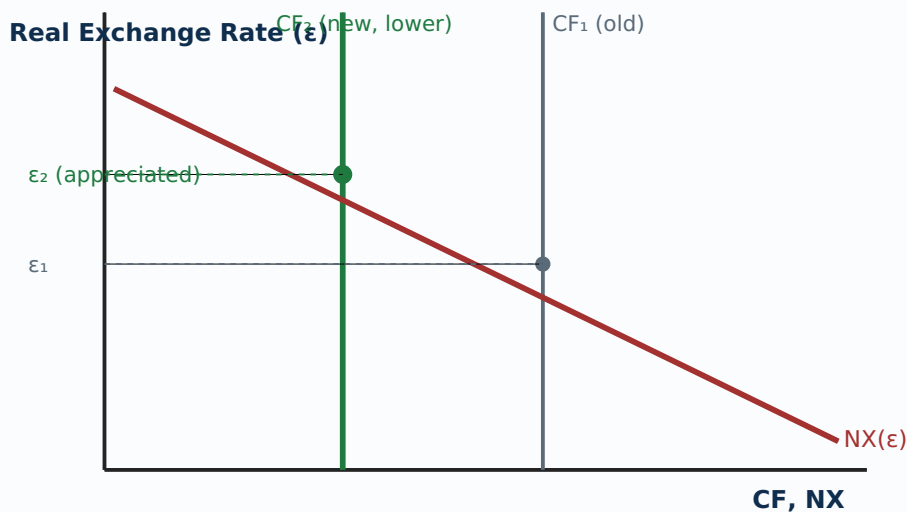


Figure 2: The market for the small economy's currency. Net capital outflow (CF, which equals net exports by identity) falls from CF_1 to CF_2 , requiring the real exchange rate to appreciate from ϵ_1 to ϵ_2 to reduce net exports to match.

The mechanism. The real exchange rate adjusts to bring net exports into line with the now-lower CF, since $CF = NX$ must hold. Net exports, $NX(\epsilon)$, is downward-sloping in the real exchange rate: a stronger (appreciated) currency makes domestic goods more expensive to foreigners and foreign goods cheaper to domestic consumers, reducing net exports. To bring NX down to match the now-lower CF, the real exchange rate must **appreciate**, from ϵ_1 to ϵ_2 in Figure 2.

Putting it together: cause and effect run entirely from the large economy's saving decision. The chain of causation is: U.S. fiscal contraction → higher U.S. national saving → lower world real interest rate → higher investment (and lower net capital outflow) in the small open economy → worsening trade balance and an appreciating real exchange rate in the small open economy, purely as a consequence of a policy decision made entirely outside its own borders. This illustrates a genuine vulnerability of small open economies: significant macroeconomic outcomes can be driven by policy changes in large economies over which the small economy has no control or influence.

How the 100% Is Likely Allocated

Tier	Marks	What separates this tier
80-100%	80-100%	Correctly traces the full causal chain (US fiscal contraction → lower world interest rate → higher small-economy investment → lower CF), correctly applies the $CF = NX$ identity to conclude the trade balance worsens, correctly derives real exchange rate appreciation from the $NX(\epsilon)$ relationship, and illustrates both steps with correctly labelled diagrams.
50-79%	50-79%	Correctly identifies the trade balance worsens and the exchange rate appreciates, but without the full loanable-funds/ $CF = NX$ mechanism, or with only one diagram.
20-49%	20-49%	Identifies one of the two effects correctly but not the other, or reasoning without diagrams.
0-19%	0-19%	No meaningful attempt, or incorrect direction for both effects.

Question A3**100%**

Ecoland is a small island economy that produces three goods. Over the past three years the economy has experienced both increases in productivity and prices. This is displayed in the table below.

Good	2019 Price	2019 Quantity	2020 Price	2020 Quantity	2021 Price	2021 Quantity
Coconuts	€3.00	60	€3.20	75	€3.30	95
Olive Oil	€3.50	65	€3.50	70	€3.60	75
Rubber	€3.80	15	€3.90	18	€4.00	25

Using this data calculate:

- (i) Nominal GDP in 2019.
- (ii) Real GDP in 2020 and 2021, using 2019 base year prices.
- (iii) The difference between Nominal GDP and Real GDP in 2021, using 2019 base year prices.

FULL MODEL ANSWER**(i) Nominal GDP in 2019**

Nominal GDP uses each year's own prices with that year's own quantities:

Good	2019 Price × 2019 Quantity	€
Coconuts	3.00×60	180.00
Olive Oil	3.50×65	227.50
Rubber	3.80×15	57.00
Nominal GDP 2019		464.50

(ii) Real GDP in 2020 and 2021, Using 2019 Base Year Prices

Real GDP holds prices fixed at the base year (2019) and applies them to each subsequent year's actual quantities — isolating the effect of changing output from the effect of changing prices.

Good	2019 Price × 2020 Quantity	€
Coconuts	3.00×75	225.00
Olive Oil	3.50×70	245.00
Rubber	3.80×18	68.40
Real GDP 2020 (2019 base prices)		538.40

Good	2019 Price × 2021 Quantity	€
Coconuts	3.00×95	285.00
Olive Oil	3.50×75	262.50
Rubber	3.80×25	95.00
Real GDP 2021 (2019 base prices)		642.50

(iii) Difference Between Nominal GDP and Real GDP in 2021

Nominal GDP in 2021 uses 2021's own (higher) prices with 2021 quantities:

Good	2021 Price × 2021 Quantity	€
Coconuts	3.30×95	313.50
Olive Oil	3.60×75	270.00
Rubber	4.00×25	100.00
Nominal GDP 2021		683.50

Difference = Nominal GDP 2021 – Real GDP 2021 (2019 base) = 683.50 – 642.50

= €41.00

This €41.00 gap between the two 2021 figures is entirely attributable to price inflation between 2019 and 2021, since both figures use the identical 2021 quantities — only the prices differ (2021's own prices vs. 2019's base-year prices). This is exactly the distinction real GDP is designed to isolate: nominal GDP growth reflects both real output growth and price inflation combined, while real GDP growth reflects output growth alone.

How the 100% Is Likely Allocated

Tier	Marks	What separates this tier
80-100%	80-100%	All calculations correct with clear working shown: Nominal GDP 2019 = €464.50; Real GDP 2020 = €538.40; Real GDP 2021 = €642.50; Nominal GDP 2021 = €683.50; difference = €41.00, with a correct explanation of why the two 2021 figures differ (price changes only, since quantities are identical in both calculations).
50-79%	50-79%	Most calculations correct with minor arithmetic slips, or the explanation of the difference in part (iii) missing.
20-49%	20-49%	Correct method for nominal GDP but real GDP calculated using the wrong year's prices (e.g. each year's own prices rather than the fixed 2019 base year prices).
0-19%	0-19%	No meaningful attempt, or a fundamentally incorrect method.

Section B — Long Questions

Answer 1 of 2 · 60% of total marks · all parts equal weight

Question B1

Productivity, Growth Theory & Institutions

B1 (a)

25%

(a) Explain the term productivity? Why does it vary across countries?

FULL MODEL ANSWER

Productivity refers to how efficiently an economy converts its inputs — physical capital and labor — into output (GDP). Formally, it is captured by the technology index, A , in the aggregate production function $Y = AF(K,H)$: for any given quantities of capital and efficiency units of labor, a higher A means more GDP is produced from the same inputs.

Why Productivity Varies Across Countries

1. Human capital. The stock of skills embodied in a country's workforce — education, training, and accumulated experience — directly affects how much output a given number of workers can produce. Countries with stronger education systems and more accumulated workforce experience have higher effective labor input ($H = L \times h$) even holding the raw number of workers constant.

2. Physical capital. The stock of machinery, equipment, and business infrastructure available to workers varies enormously across countries. More and better capital gives workers more tools to work with, directly raising output per worker — though, as the aggregate production function's diminishing marginal product property shows, this effect diminishes as the capital stock grows relative to the workforce.

3. Technology. Beyond the quantities of capital and labor themselves, countries differ in the superior knowledge and efficiency of their production processes — represented by the technology index A itself. A country with more advanced technology produces more output from identical quantities of capital and labor than a country with less advanced technology; this is the specific channel through which innovation, know-how, and managerial practice affect productivity independent of factor accumulation.

A deeper explanation: institutions shape all three proximate sources.

Beneath these three proximate sources lies a deeper explanation, discussed further in part (d): a country's institutions — property rights, rule of law, political stability — shape the incentives to accumulate human capital (why invest in education if returns aren't secure?), physical capital (why invest if property rights are weak?), and technology (why innovate if the gains can be expropriated?) in the first place. Institutions are therefore often described as the fundamental cause of productivity differences, with human capital, physical capital, and technology as the proximate (more immediately visible) causes.

How the 25% Is Likely Allocated

Tier	Marks	What separates this tier
20-25%	20-25%	Correctly defines productivity with reference to the aggregate production function (the role of A), correctly identifies all three proximate sources (human capital, physical capital, technology) with explanation of each, and connects to the deeper institutional explanation.
12-19%	12-19%	Correctly identifies the three proximate sources with reasonable explanation, but without the deeper institutional connection.
5-11%	5-11%	Vague definition of productivity, or only one or two sources identified.
0-4%	0-4%	No meaningful attempt.

B1 (b)**25%**

(b) Illustrate the Solow growth model to answer the following questions.

- i. What is the impact of a higher saving rate on GDP? Explain the process.
- ii. What is the impact of an increase in labor force on GDP? Explain the process.

FULL MODEL ANSWER

The Solow model tracks capital per worker (k) on the horizontal axis against output and investment per worker on the vertical axis. Output per worker, $y = f(k)$, is the familiar concave production function; actual investment per worker is $sf(k)$, a constant fraction s of output; and break-even investment, $(n+d)k$, is the amount of investment needed just to maintain the existing capital-per-worker ratio given population growth (n) and depreciation (d). The steady state, k^* , occurs where these two lines intersect — capital per worker (and therefore output per worker) is stable, neither rising nor falling.

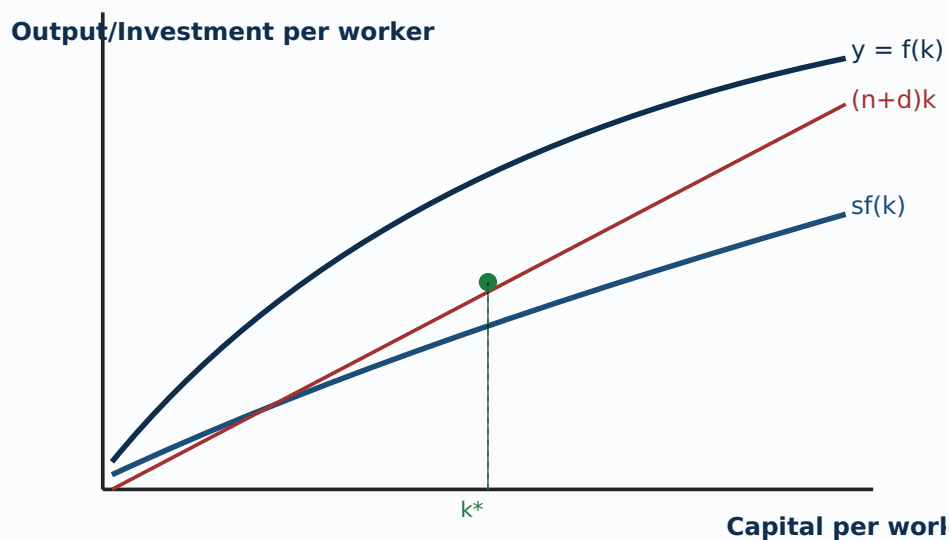


Figure 1: The Solow growth model. Steady-state capital per worker (k^*) occurs where investment, $sf(k)$, exactly equals break-even investment, $(n+d)k$ — the amount needed to replace depreciating capital (d) and equip new workers (n) at the existing capital-per-worker ratio.

(i) Impact of a Higher Saving Rate on GDP

The process: a higher saving rate shifts the investment curve upward.

An increase in the saving rate, s , shifts the $sf(k)$ curve upward at every level of k (since a larger fraction of output is now invested rather than consumed). At the original steady state k^* , actual investment now exceeds break-even investment, so capital per worker begins to rise.

The economy transitions to a new, higher steady state. As k rises, output per worker $y = f(k)$ rises with it, though at a diminishing rate (the concave production function). Capital per worker keeps rising until it reaches a new steady state, k^{**} , at a higher level than before, where the new (higher) $sf(k)$ curve again intersects the unchanged $(n+d)k$ line. GDP per worker is therefore permanently higher in the new steady state — but importantly, this is a level effect, not a permanent growth-rate effect: once the new steady state is reached, GDP per worker stops growing again (absent further increases in s or improvements in technology), since diminishing returns to capital mean saving alone cannot sustain perpetual growth.

(ii) Impact of an Increase in the Labor Force on GDP

The process: population/labor force growth raises the break-even investment line. A higher rate of labor force growth, n , increases the $(n+d)k$ line's slope — more new investment is now needed just to equip the growing number of new workers with the existing amount of capital per worker, on top of replacing depreciation. At the original steady state, break-even investment now exceeds actual investment $sf(k)$, so capital per worker begins to fall.

The steady state shifts to a lower level of capital (and output) per worker. Capital per worker falls until a new, lower steady state is reached, where the steeper $(n+d)k$ line intersects the unchanged $sf(k)$ curve. GDP per worker is therefore lower in the new steady state than before — even though total GDP (not per worker) may still rise, since there are now more workers overall, each producing less than before due to more of them sharing the same capital stock (a lower capital-to-labor ratio).

This distinguishes total GDP growth from GDP per worker/capita growth. A larger labor force mechanically raises total output (more workers producing something), but the model shows that rapid population growth can actually reduce living standards (GDP per worker), since capital has to be spread more thinly — a key reason cross-country comparisons of prosperity focus on GDP per capita rather than total GDP.

How the 25% Is Likely Allocated

Tier	Marks	What separates this tier
20-25%	20-25%	Correctly draws/describes the Solow diagram with all three curves labelled, correctly explains the saving rate increase as an upward shift in $sf(k)$ leading to a higher steady state (with the level-vs-growth-rate distinction noted), and correctly explains labor force growth as steepening $(n+d)k$ leading to a lower steady-state capital/output per worker (distinguishing total GDP from GDP per worker).
12-19%	12-19%	Correctly identifies the direction of both effects but with less precise explanation of which curve shifts and why.
5-11%	5-11%	One of the two sub-questions answered correctly; the other vague or missing.
0-4%	0-4%	No meaningful attempt, or no reference to the Solow model's actual structure.

B1 (c)**25%**

(c) What did Malthus predict about economic growth? Did his predictions come true? Why or why not?

FULL MODEL ANSWER

Malthus's Prediction

Humankind was destined to remain at subsistence level, indefinitely.

Writing in 1798, Thomas Malthus argued that any rise in income per capita above the minimum needed for survival (subsistence level) would trigger higher fertility rates. The resulting population growth would then outstrip the (largely fixed, technologically static) capacity of land to produce food, driving income per capita back down to subsistence. Under this logic, sustained improvement in living standards was essentially impossible — any gains would always be reabsorbed by population growth, trapping humanity in a repeating cycle now known as the Malthusian Trap.

Did His Predictions Come True?

True for most of history, but decisively wrong from the eighteenth century onward

The prediction held remarkably well for the vast majority of human history before Malthus wrote. For roughly 2,800 years before 1800, global GDP per capita showed essentially no sustained growth, consistent with economies cycling between temporary improvements and population-driven reversals exactly as Malthus described — strong historical support for the theory as a description of the pre-modern world.

But the prediction failed dramatically from around the mid-eighteenth century onward. Sustained, compounding growth in income per capita began in Britain and then spread globally, with income per capita reaching roughly 12 times its 1750 level by the year 2000 — the exact opposite of a world permanently trapped at subsistence.

Why the Predictions Ultimately Failed

Malthus's model implicitly assumed technology and fertility behaviour would stay fixed. The Malthusian cycle only holds if food-producing technology stays essentially static (so population growth genuinely does outrun the food supply) and if fertility keeps rising whenever income rises above subsistence. Two developments broke both assumptions.

First, the Industrial Revolution created sustained, self-reinforcing technological progress. New machines and production methods, beginning in textile manufacturing and spreading through the economy, allowed output to grow faster than population for the first time in a durable, ongoing way — unlike earlier isolated bursts of innovation, each advance enabled further advances rather than petering out.

Second, the demographic transition broke the assumed fertility response to rising income. As populations urbanised and moved from agriculture into manufacturing, fertility rates fell rather than rose as incomes increased — children shifted from productive farm labour to a costly, education-intensive investment, reducing the incentive for large families. This is precisely the opposite of what Malthus's model assumed would happen as income rose.

Together, these two changes let rising income actually persist, rather than being reabsorbed by population growth. It's the *combination* of technological progress and falling fertility that explains the escape from the Malthusian Trap — technological progress alone, without the accompanying fertility change, would likely have simply been absorbed by a larger population, exactly as it always had been in every earlier episode of innovation before the eighteenth century.

How the 25% Is Likely Allocated

Tier	Marks	What separates this tier
20-25%	20-25%	Correctly explains the Malthusian Trap mechanism, correctly notes the prediction held for most of pre-1800 history but failed thereafter, and explains both reasons for the failure (Industrial Revolution technology AND the demographic transition), with the interaction between them made explicit.
12-19%	12-19%	Correctly explains the Malthusian mechanism and identifies the Industrial Revolution as the reason it failed, but without the demographic transition/fertility dimension.
5-11%	5-11%	Vague or partial description of Malthus's prediction without a clear explanation of why it failed.
0-4%	0-4%	No meaningful attempt.

B1 (d)**25%**

(d) As defined by economic historian Douglass North, (1990) “institutions are the rules of the game in a society or, more formally, are the humanly devised constraints that shape human interaction.”

i. Does the role of institutions or geography offer a more plausible explanation of cross-country differences in economic growth? Explain with reference to the work of Daron Acemoglu.

ii. How does the institution hypothesis explain the difference, in terms of economic growth, between North and South Korea?

FULL MODEL ANSWER**(i) Institutions vs. Geography****Institutions offer the more plausible explanation**

The geography hypothesis, and its limitations. Geographic theories attribute cross-country income differences to climate, natural resources, disease burden, and access to trade routes — tropical regions, for instance, have historically faced worse agricultural productivity and higher disease burdens than temperate regions, which some scholars (such as Jeffrey Sachs) argue plays a first-order role in explaining persistent poverty. Geography clearly matters at the margin, but it struggles to explain cases where genetically, climatically, and geographically near-identical populations end up with vastly different economic outcomes purely because of a border.

Acemoglu's reversal-of-fortune evidence directly challenges the geography hypothesis. Daron Acemoglu, alongside Simon Johnson and James Robinson, documented that regions which were relatively prosperous 500 years ago (before European colonisation) are frequently relatively poor today, and vice versa — a “reversal of fortune” that is very difficult to explain through geography alone, since a region's climate and natural endowments haven't changed over that period, but its relative prosperity has reversed. Acemoglu and Robinson argue the explanation lies in the type of institutions European colonisers established: in regions with a high disease burden for European settlers, colonisers tended to install extractive institutions (designed to extract resources with minimal investment in long-run development); in regions more hospitable to European settlement, colonisers built inclusive institutions resembling those of their home countries (secure property rights, broadly distributed political power) — and it is this institutional difference, not the underlying geography, that best predicts subsequent economic performance.

Twin-city and divided-country natural experiments make the same point even more directly. Comparisons like Nogales (a single city split by the US-Mexico border) or North versus South Korea (below) hold geography, climate, and ancestry constant while institutions diverge sharply — and in every such case, it is the institutional divergence, not any geographic difference (since there is none), that tracks the resulting gap in prosperity.

(ii) North Korea vs. South Korea

A near-perfect natural experiment for isolating institutions from every other factor. Before their division following the Second World War, the Korean peninsula was a single people sharing identical geography, climate, culture, language, and history. Since 1948, the two halves have followed dramatically different institutional paths: South Korea developed increasingly inclusive economic and political institutions (secure property rights, market-based resource allocation, and eventually democratic political institutions), while North Korea adopted a centrally planned, extractive economic system under an authoritarian political regime with no meaningful checks on power.

The economic outcomes track the institutional divergence almost exactly. South Korea transformed from one of the poorest countries in the world in the 1950s into a high-income, technologically advanced economy, while North Korea has remained impoverished, with satellite imagery of the peninsula at night showing North Korea almost entirely dark compared to a brightly lit South — a visually stark illustration of the underlying economic gap. Because every other plausible explanatory factor (geography, culture, initial history) is held constant between the two, the institution hypothesis identifies this divergent economic performance as attributable specifically to the different economic and political institutions each half subsequently adopted, not to anything that predates the division.

How the 25% Is Likely Allocated

Tier	Marks	What separates this tier
20-25%	20-25%	Takes a clear position favouring institutions, correctly explains Acemoglu's reversal-of-fortune evidence and the extractive-vs-inclusive institutions framework, AND correctly explains why North/South Korea is such a clean test of the institution hypothesis (geography/culture/history held constant, institutions vary, outcomes track institutions).
12-19%	12-19%	Correctly favours institutions with a reasonable explanation, but without the specific Acemoglu reversal-of-fortune evidence or with a less complete Korea explanation.
5-11%	5-11%	Vague preference for institutions or geography without engaging with the Acemoglu literature or the Korea comparison specifically.
0-4%	0-4%	No meaningful attempt.

Question B1 Total: 100% (60% of the Paper)

Question B2

Central Banks & European Monetary Union

B2 (a)



(a) Briefly explain two primary functions of central banks such as the Federal Reserve or Bank of England.

FULL MODEL ANSWER

1. Conducting monetary policy to control inflation and support employment/growth. Central banks set short-term interest rates (or, more directly, control the money supply) to influence borrowing costs, spending, investment, and ultimately inflation and output across the economy. This is the central bank's most visible function: raising rates to cool an overheating, inflationary economy, or cutting rates (and in extreme cases undertaking quantitative easing) to stimulate a weak economy, typically guided by an explicit mandate (such as an inflation target).

2. Acting as lender of last resort and safeguarding financial stability. Central banks stand ready to lend to solvent but temporarily illiquid banks during a financial panic, preventing a liquidity shortage at one institution from cascading into a systemic banking crisis through contagion and bank runs. This function — alongside broader financial system oversight and regulation — is what distinguishes a central bank from an ordinary commercial bank, and was exercised prominently by both the Federal Reserve and the Bank of England during the 2008 financial crisis.

Other valid functions include issuing the national currency (banknotes) and acting as banker to the government (managing government accounts and debt issuance) — either would also be creditable as one of the two functions, provided it's clearly explained.

Marking Guidance

Tier	Marks	What separates this tier
Full credit	Full credit	Two genuinely distinct central bank functions identified (from monetary policy, lender of last resort/financial stability, currency issuance, or government banker) with a clear explanation of what each involves.
Partial credit	Partial credit	One function well explained; the second vague or missing.
Minimal/no credit	Minimal/no credit	No meaningful attempt, or functions named with no explanation.

B2 (b)

(b) The Delors Report and Maastricht Treaty paved the way for the creation of the single currency (the euro).

With reference to both, explain three of the five 'strict economic criteria' required for successful operation of the currency and the European Central Bank (ECB).

FULL MODEL ANSWER

The 1989 Delors Report set out the roadmap for European Economic and Monetary Union in three stages, culminating in a single currency; the 1992 Maastricht Treaty then formalised this into binding law, including five specific “convergence criteria” each member state must meet before adopting the euro — designed to ensure entering economies were sufficiently aligned that a single monetary policy set by the ECB could sensibly apply to all of them at once. Three of the five criteria:

1. Price Stability (Inflation Criterion)

The requirement. A candidate country's inflation rate, measured over the year before assessment, must not exceed the average inflation rate of the three best-performing (lowest-inflation) EU member states by more than 1.5 percentage points.

Why it matters for successful monetary union. Since the ECB sets a single interest rate for the entire eurozone, a country entering with a significantly higher inflation rate than its partners would need a persistently different (typically higher) interest rate than the single ECB rate could ever provide, creating an immediate structural mismatch between that country's needs and the currency union's shared monetary policy.

2. Sound Public Finances (Government Deficit and Debt Criteria)

The requirement. A candidate country's annual government budget deficit must not exceed 3% of GDP, and its total government debt must not exceed 60% of GDP (this is sometimes counted as two separate criteria — deficit and debt — making up the “fiscal criterion,” which is why the full framework is sometimes described as having five components across four broad areas).

Why it matters for successful monetary union. Members of a currency union no longer have their own currency to devalue or their own central bank to monetise excessive debt, so a fiscally undisciplined member risks a genuine sovereign debt crisis that can spill over and threaten the stability of the whole currency union — exactly what happened during the eurozone sovereign debt crisis beginning in 2010, when Ireland and several other member states saw their borrowing costs spike sharply as markets questioned fiscal sustainability.

3. Exchange Rate Stability

The requirement. A candidate country must participate in the Exchange Rate Mechanism (ERM II) for at least two years without severe tensions, keeping its currency within agreed fluctuation margins against the euro and without devaluing against any other member state's currency during that period.

Why it matters for successful monetary union. Demonstrating a sustained period of exchange rate stability against the euro, prior to fully adopting it, is intended to show the candidate's economy is genuinely converged with the eurozone rather than only artificially or temporarily aligned, reducing the risk of needing a disruptive one-off exchange rate adjustment immediately after joining.

The remaining two criteria not detailed above are the long-term interest rate criterion (a country's long-term government bond yield must not exceed that of the three best-performing member states by more than 2 percentage points, similarly ensuring genuine convergence in market-assessed creditworthiness) and legal convergence (national legislation, including the country's own central bank statute, must be compatible with the EU treaties and the ECB's independence) — either would also be creditable as one of the three required criteria.

Marking Guidance

Tier	Marks	What separates this tier
Full credit	Full credit	Three of the five criteria correctly stated with accurate thresholds (inflation within 1.5 points of the three best performers; deficit $\leq 3\%$ of GDP and debt $\leq 60\%$ of GDP; ERM II participation for 2+ years; interest rates within 2 points of the three best performers; legal convergence), each with an explanation of why it matters for a functioning currency union.
Partial credit	Partial credit	Three criteria named but with inaccurate thresholds, or correct criteria with limited explanation of their rationale.
Minimal/no credit	Minimal/no credit	Fewer than three criteria identified, or no reference to the actual Maastricht framework.

B2 (c)

(c) Explain two economic benefits and one economic cost to Ireland from being a member of the European Monetary Union (EMU).

FULL MODEL ANSWER**Benefit 1: Eliminated Exchange Rate Risk and Transaction Costs with Eurozone Trading Partners**

The mechanism. Ireland conducts a very large share of its trade with other eurozone members; sharing a single currency with them eliminates the exchange rate risk and currency conversion costs that would otherwise apply to every cross-border transaction. This gives Irish exporters and importers greater price certainty and removes a real cost that firms trading across different currencies must otherwise bear (hedging costs, conversion fees, and the risk of adverse currency movements between agreeing a price and being paid).

Benefit 2: Enhanced Credibility, Lower Borrowing Costs, and Access to a Deep Capital Market

The mechanism. As a small economy, Ireland benefits from the euro's status as a major global reserve and trading currency, backed by the credibility of the European Central Bank's institutional commitment to price stability — a credibility a small, standalone Irish currency would have struggled to establish independently. Eurozone membership also gives Irish borrowers (government and, indirectly, firms) access to a much deeper and more liquid pool of capital than a small national currency and bond market could offer on its own, generally supporting lower borrowing costs in normal times.

Cost: Loss of Independent Monetary Policy

The mechanism. Ireland no longer has its own interest rate or exchange rate to adjust in response to conditions specific to the Irish economy — monetary policy is set by the ECB for the eurozone as a whole, which can be a poor fit for Ireland's own specific economic conditions at any given time (the “one size fits all” problem). ECB interest rates during the mid-2000s, appropriate for a broadly sluggish eurozone average, were arguably too low for an Irish economy running well above its sustainable trend during the Celtic Tiger boom, contributing to the property bubble that preceded Ireland's own financial crisis. When an asymmetric shock hits Ireland specifically (one not affecting the rest of the eurozone equally), Ireland can no longer respond with its own interest rate cut or a currency devaluation — fiscal policy is left as essentially the only national-level lever available, as discussed further in Question B2(d) of the related paper on Irish fiscal and monetary policy.

Other valid benefits include increased price transparency (making cross-border price comparison easier for consumers and firms, intensifying competition) and greater foreign direct investment attractiveness (multinationals can serve the eurozone market from an Irish base without currency conversion friction) — either would also be creditable as a second benefit. Other valid costs include the loss of the exchange rate as a shock-absorption tool specifically (distinct from, though related to, the loss of monetary policy more broadly) and reduced fiscal policy flexibility due to EU-level fiscal rules that accompany EMU membership.

Marking Guidance

Tier	Marks	What separates this tier
Full credit	Full credit	Two genuinely distinct economic benefits AND one economic cost identified, each explained with a clear mechanism specific to Ireland's situation as a small open economy within the eurozone, ideally with a concrete illustration (such as the Celtic Tiger interest rate mismatch).
Partial credit	Partial credit	Benefits and cost named but with limited explanation of the underlying mechanism, or generic points not specifically tied to Ireland.
Minimal/no credit	Minimal/no credit	No meaningful attempt, or fewer than the required number of benefits/costs identified.

Question B2 Total: 100% (60% of the Paper)